# AN EXPLICIT 13-STAGE RUNGE-KUTTA METHOD OF ORDER 9 WITH EXACT ALGEBRAIC COEFFICIENTS

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Abstract. We give an explicit 13-stage Runge-Kutta method of order 9 whose coefficients are real algebraic numbers in a totally real number field of degree 16, and we verify all 486 order conditions in exact rational arithmetic. This proves that the minimal number of stages $a(9)$ of an explicit Runge-Kutta method of order 9 satisfies $a(9) \le 13$. Together with Butcher's lower bound $s \ge p + 3$ for order $p \ge 8$, this gives $12 \le a(9) \le 13$ (OEIS A187102). The method is not the 13-stage method that Khashin found numerically in 2009. It lies on a second five-dimensional component of the solution set, on which four weights and eight coupling coefficients vanish, three nodes coincide, and the weights form the interpolatory quadrature rule of nine nodes. Four programs, from three separate sources, confirm the exact check; one of them uses only the integer arithmetic of a stock Python interpreter and runs in about a minute. The search that produced the method, which ends in a 25000-digit refinement and a lattice reconstruction, is described but is not part of the proof.

## 1. Introduction

An explicit Runge-Kutta method with $s$ stages is given by a strictly lower triangular $s \times s$ matrix $A = (a_{ij})$ and a weight vector $b = (b_1, \dots, b_s)$, with nodes $c_i = \sum_j a_{ij}$. The method has order $p$ when its coefficients satisfy one polynomial equation for every rooted tree with at most $p$ vertices [4, 8]. Let $a(p)$ be the smallest $s$ for which an explicit method of order $p$ exists. The values $a(1), \dots, a(8)$ are

$$1,\ 2,\ 3,\ 4,\ 6,\ 7,\ 9,\ 11,$$

and they are listed as OEIS sequence A187102 [18]. The upper bounds come from explicit constructions; for order 8 with 11 stages see Cooper and Verner [5]. The lower bounds are Butcher's barrier theorems: $s \ge p + 1$ for $p \ge 5$, $s \ge p + 2$ for $p \ge 7$, and $s \ge p + 3$ for $p \ge 8$ [1, 2, 3]; see the summary in [27, Theorem 1.27]. So order 9 needs at least 12 stages.

For order 9 the best upper bound has been numerical for seventeen years. Khashin [12, 13] found a 13-stage method of order 9 by a Tikhonov-regularized Newton iteration on a system reduced by what he calls filtrated simplifying assumptions, and printed its tableau to 17 decimal places. His own wording is that the methods "has been obtained (at the moment numerically only)" [13, Section 4.4]. Before that, the best known stage count for order 9 was 15 [24], as Khashin notes; later surveys cite his 13-stage method as a numerical result [26, 27]. The OEIS entry, read on 2026-10-30, says: "Khashin (2009) numerically found a 13-stage method of order 9, so it seems that $a(9) \le 13$" [18]. Zhang writes that Khashin's method "reduces the uncertainty for $p = 9$ to a single stage $12 \le s \le 13$" [27], and Ketcheson, right after noting that the coefficients of Stepanov's 15-stage order-10 methods are known exactly, writes that Khashin's method "is not known to as many digits of accuracy", but that "most experts believe that 13-stage methods of order 9 exist" [10]. For order 10, explicit methods are known with 18 stages [6], 17 stages [7], 16 stages found numerically [28], and 15 stages with exact coefficients [20], so $13 \le a(10) \le 15$.

This paper turns the order-9 bound into a theorem.

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**Theorem 1.1.** *There is an explicit 13-stage Runge-Kutta method of order 9 with real coefficients. More precisely, there is a totally real number field $K$ of degree 16 over $\mathbb{Q}$ and a tableau $(A, b)$ with entries in $K$ that satisfies all 486 order conditions of order 9 exactly in $K$. Each of the 16 real embeddings of $K$ gives a real explicit 13-stage method of order 9.*

**Corollary 1.2.** $12 \le a(9) \le 13$.

The proof is a finite exact computation: the coefficients are given as explicit elements of a ring $\mathbb{Q}[t, y]/(F(t), y^2 - P_1(t)\, y - P_0(t))$, with $F$ of degree 8 and $P_0, P_1$ of degree 7, and the 486 order conditions, which are polynomial identities in the coefficients with rational constants, are reduced to zero in that ring. A real root of $F$ at which the quadratic in $y$ has real roots then gives a real method. Nothing in the proof depends on floating-point arithmetic or on how the coefficients were found. The existence of the 16 real methods rests only on exact rational arithmetic, and four programs from three separate sources agree on it, one of them written only with Python's built-in integers and its `fractions` module. The statement that $K$ is a field of degree 16 uses in addition the factorization of polynomials over $\mathbb{Q}$ in FLINT, done in two separately written scripts. Section 4 states the lemmas and the checks.

The method is not Khashin's. His tableau refines, by Gauss-Newton iteration in high precision, to a point that satisfies the order conditions to $1.1 \times 10^{-223}$ with quadratic convergence, which is strong numerical evidence that his method is of order 9, but we have no exact certificate for it. The solutions near his tableau form a smooth five-dimensional family $G$. Following $G$ to the hypersurface $b_3 = 0$ led to a second five-dimensional component $\Sigma$ of the solution set, with more vanishing coefficients and more structure, which meets $G$ in a four-dimensional junction. On $\Sigma$ it was possible to fix five parameters at rational values so that every remaining coefficient lies in a number field of degree 16. Section 5 describes this search. The question whether $a(9) = 12$ or 13 remains open, as does the exact status of Khashin's tableau.

To our knowledge, no explicit 13-stage method of order 9 with exact coefficients, and no proof that one exists, has been published before. On 2026-10-30 we searched arXiv, the OEIS history, MathOverflow, GitHub, Zenodo, the general web and the pages of Khashin and Verner (Verner's page returned an error, and archived copies were read); Feagin's page did not respond. On 2026-10-31 and again on 2026-11-01 we repeated the arXiv, OEIS, GitHub and web searches.

## 2. Order conditions

We use Butcher's rooted-tree formulation [4, 8], in the form in which the checking programs implement it. A rooted tree is either the single vertex $\tau$ or $[t_1, \dots, t_k]$, obtained by joining the roots of the trees $t_1, \dots, t_k$ to a new root; the order of the $t_m$ does not matter. The number of vertices is $|\tau| = 1$ and $|[t_1, \dots, t_k]| = 1 + \sum_m |t_m|$. The numbers of rooted trees with $n = 1, \dots, 9$ vertices are

$$1,\ 1,\ 2,\ 4,\ 9,\ 20,\ 48,\ 115,\ 286$$

(OEIS A000081 [19]), and there are 486 trees with at most 9 vertices.

For an $s$-stage tableau $(A, b)$ the stage vectors $\Phi(t) \in \mathbb{R}^s$ (or in any commutative ring containing the entries) are defined recursively by

$$\Phi(\tau) = \mathbf{1}, \qquad \Phi([t_1, \dots, t_k]) = \prod_{m=1}^{k} A\, \Phi(t_m),$$

where $\mathbf{1}$ is the vector of ones and the product of vectors is taken componentwise. The density is $\gamma(\tau) = 1$ and $\gamma([t_1, \dots, t_k]) = |t| \prod_m \gamma(t_m)$. With the convention $c = A\mathbf{1}$, the method has order

$p$ if and only if

$$b \cdot \Phi(t) = \frac{1}{\gamma(t)} \qquad \text{for every rooted tree } t \text{ with } |t| \le p. \tag{1}$$

This is Butcher's theorem [4, Section 315], [8, p. 153]. Order 9 is the system of 486 equations (1) for $|t| \le 9$. Each $\Phi_i(t)$ is a polynomial with integer coefficients in the entries of $A$, of degree $|t|-1$, and $b \cdot \Phi(t)$ is a polynomial of degree $|t|$ in the entries of $(A, b)$. The nodes do not appear as separate unknowns: a bushy tree $[\tau, \dots, \tau]$ with $k$ leaves gives $\Phi = c^k$ (componentwise power) and the condition $b \cdot c^k = 1/(k+1)$, which for $k = 0, \dots, 8$ is the quadrature condition $B(9)$ in Butcher's notation.

Throughout, $s = 13$ and $p = 9$. The unknowns are the 78 entries $a_{ij}$, $1 \le j < i \le 13$, and the 13 weights $b_i$, in all 91 numbers.

## 3. The method

3.1. **Shape.** Fourteen of the 91 coefficients are rational and are fixed at the following values:

$$\begin{gathered} a_{5,2} = a_{6,2} = a_{7,2} = a_{8,2} = a_{10,2} = a_{11,2} = a_{12,2} = a_{13,2} = 0, \\ b_2 = b_3 = b_4 = b_9 = 0, \qquad a_{2,1} = \tfrac{5}{4}, \qquad a_{13,9} = -4. \end{gathered} \tag{2}$$

So stage 2 feeds only stages 3, 4 and 9 (the nonzero entries of column 2 of $A$ are $a_{3,2}$, $a_{4,2}$ and $a_{9,2}$), and only nine stages carry a nonzero weight: stages 1, 5, 6, 7, 8, 10, 11, 12 and 13. Figure 1 shows the pattern. The other 77 coefficients are the free coordinates of the certificate. They are algebraic numbers, described in Section 3.3, and the entries $a_{4,2}$ and $a_{9,2}$ coincide (their stored forms in the data file are identical).

The two values $a_{2,1} = 5/4$ and $a_{13,9} = -4$, and the three node values below, are the rational parameters at which the five-dimensional component $\Sigma$ of Section 5 was cut down to a finite set of points; they were chosen for the size of the resulting algebraic numbers, not for the quality of the method. In particular $c_2 = 5/4$ lies outside $[0, 1]$. (Khashin's tableau has $c_2 = 1.23456789012345679$, which shows that this node is a free parameter of his family too; the corresponding one-parameter deformation is his Proposition 3.11 [13].)

3.2. **Nodes and weights.** Seven of the thirteen nodes are rational:

$$c_1 = 0, \quad c_2 = \tfrac{5}{4}, \quad c_4 = c_9 = c_{10} = \tfrac{6}{25}, \quad c_5 = \tfrac{187}{250}, \quad c_{11} = \tfrac{3}{4}, \quad c_{13} = 1.$$

The nodes $c_3, c_6, c_7, c_8, c_{12}$ are algebraic of degrees $8, 4, 4, 2, 4$ (Table 1). The nine weighted stages have the nine nodes

$$0,\ c_5,\ c_6,\ c_7,\ c_8,\ c_4,\ c_{11},\ c_{12},\ 1,$$

and the weights are determined by them.

**Lemma 3.1.** *The nine nodes $0, c_5, c_6, c_7, c_8, c_4, c_{11}, c_{12}, 1$ are pairwise distinct elements of $K$, and $b$ is the interpolatory quadrature rule of these nodes on $[0, 1]$: the nine nonzero weights are the unique solution of $\sum_i b_i c_i^k = 1/(k+1)$, $k = 0, \dots, 8$.*

*Proof.* The bushy trees with $k$ leaves give the conditions $b \cdot c^k = 1/(k+1)$ for $k = 0, \dots, 8$, which hold exactly in $K$ by Theorem 4.1, and $b$ is supported on the nine stages with these nodes. For distinctness, $K$ is a field (Section 3.3, from the irreducibility of $f$; the proof of Theorem 1.1 in Section 4 repeats the argument), so a difference of two nodes is either zero or a nonzero element of $K$, and a nonzero element of a field is nonzero under every embedding; at the embedding of Table 2 the nine nodes take the values 0, 0.748, 0.3479…, 0.9134…, 0.1450…, 0.24, 0.75, 0.4208…, 1, which are distinct. Hence the $9 \times 9$ Vandermonde matrix of the nodes is invertible over $K$ and the nine conditions determine the nine weights. □

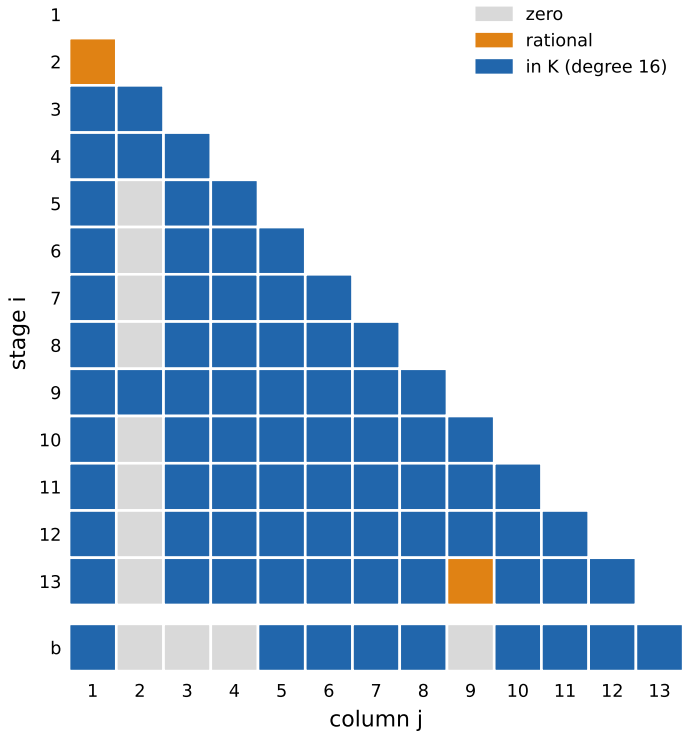



FIGURE 1. The pattern of the tableau. Rows are the stages $i = 1, \dots, 13$ and columns $j = 1, \dots, 12$ of $A$; the row of weights $b$ is below. Entries with $j \geq i$ are structurally zero, since the method is explicit. Of the 78 couplings and 13 weights, 12 are zero by (2): the eight entries $a_{i,2}$ with $i \in \{5, 6, 7, 8, 10, 11, 12, 13\}$ and the weights $b_2, b_3, b_4, b_9$. Two are rational, $a_{2,1} = 5/4$ and $a_{13,9} = -4$. The remaining 77 entries, 68 couplings and 9 weights, lie in the field $K$ of degree 16. Column 2 is nonzero only in rows 3, 4 and 9. Grey: zero; orange: rational; blue: in $K$.

The weights are not all positive: at the embedding of Table 2, $b_{10} \approx -0.0741$ and $b_{11} \approx -6.544$ are negative, and $b_5 \approx 6.789$ is the largest weight.

3.3. **The field.** Let $t = c_3$ and $y = c_{12}$. The certificate presents the coefficients in two equivalent forms.

*Tower form.* $t$ is a root of an irreducible polynomial $F \in \mathbb{Z}[t]$ of degree 8 with integer coefficients of up to 72 digits, all of whose 8 roots are real:

| $k$ | $F_k$ |
|---|---|
| 0 | 38365330100392129187408802447253381210555239661907715732556627842816 |
| 1 | -98259301196816873789470608310398768249998387377793883416023428988646 4 |
| 2 | 10354943678377134543803686089059595736461871483291505638236026410986912 |
| 3 | -57913324738371110890419897535473040267266954989805903067604464172126160 |
| 4 | 18560089331782089186127799503096887723208105585141718899102326295903533 5 |
| 5 | -34637653209345159394091459608998991684532847628244875873765667905362000 0 |
| 6 | 37041374661557417490057497943935661841127328941401221563773786486500000 0 |
| 7 | -21022359625878600629766375545766121423408611954935196686273013000000000 0 |
| 8 | 49092366502333446490590833548640662626969914262450408644085000000000000 |

The table lists the coefficients $F_k$ of $F(t) = \sum_{k=0}^{8} F_k t^k$, each split over two lines. Over $\mathbb{Q}(t)$, $y$ satisfies the quadratic $y^2 = P_1(t)\,y + P_0(t)$ with $P_0, P_1 \in \mathbb{Q}[t]$ of degree 7. Every free coefficient is written as $u(t) + v(t)\,y$ with $u, v \in \mathbb{Q}[t]$ of degree at most 7; the rational coefficients of the $u$ and $v$ have numerators of up to 872 digits and denominators of up to 862 digits. We set

$$R = \mathbb{Q}[t, y]/\big(F(t),\ y^2 - P_1(t)\,y - P_0(t)\big).$$

As a $\mathbb{Q}$-vector space $R$ has the basis $t^i y^j$, $0 \le i \le 7$, $0 \le j \le 1$, of dimension 16, and every element has a unique reduced form in this basis.

*Univariate form.* Let $\theta = c_{12} + c_3 = y + t$ and let $f \in \mathbb{Q}[T]$ be its minimal polynomial: $f$ is monic, squarefree and irreducible of degree 16, with rational coefficients whose numerators have up to 434 digits and denominators up to 430. Every free coefficient is $x_j = g_j(\theta)/f'(\theta)$ with $g_j \in \mathbb{Q}[T]$ (numerators of up to 1202 digits, denominators of up to 1196). All 16 roots of $f$ are real, with certified rational isolating intervals.

The two forms describe the same field $K = \mathbb{Q}(t, y) = \mathbb{Q}(\theta)$. That $[K : \mathbb{Q}] = 16$ follows from the irreducibility of $f$ (checked by factorization over $\mathbb{Q}$ in FLINT [21]); the independent checker also found that the characteristic polynomial of $y + kt$ acting on $R$ is irreducible of degree 16 for $k = 1, 2, 3, -1$, and that the monic characteristic polynomial of $y + t$ equals $f$. Since all 16 roots of $f$ are real, $K$ is totally real.

$K$ is a tower of quadratic extensions,

$$\mathbb{Q} \subset \mathbb{Q}(c_8) \subset \mathbb{Q}(c_7) \subset \mathbb{Q}(c_3) \subset K = \mathbb{Q}(c_3, c_{12}),$$

each of degree 2 over the one before, with $\mathbb{Q}(c_6) = \mathbb{Q}(c_7)$. The node $c_{12}$ is quadratic over $\mathbb{Q}(c_8)$ but does not lie in $\mathbb{Q}(c_3)$. So every coefficient of the method can be written with square roots nested to depth at most 4. The proof uses the exact minimal polynomials of the nodes, obtained by factoring the characteristic polynomial of multiplication by each node on $R$, and Trager's norm criterion [23]: if $\alpha$ and $\beta$ have minimal polynomials $g_\alpha$ and $g_\beta$ and the norm $N_k(z) = \prod_{i,j}(z - k\alpha_i - \beta_j)$ over their conjugates is squarefree, then the irreducible factors of $g_\beta$ over $\mathbb{Q}(\alpha)$ of degree $e$ correspond to the irreducible factors of $N_k$ of degree $e \deg g_\alpha$. The norms were computed exactly from power sums and factored in FLINT (`subfield_test.py`). Of the 77 free coordinates, exactly 31 lie in $\mathbb{Q}(c_3)$ ($v = 0$ in the tower form): every free entry of stages 3 to 8, $a_{9,2}$, and the free entries of row 11 other than $a_{11,4}$ and $a_{11,9}$. None of this is used in the proof.

The real points of $R$ are the pairs $(t_0, y_0)$ with $F(t_0) = 0$ and $y_0^2 = P_1(t_0)y_0 + P_0(t_0)$. The discriminant $D(t) = P_1(t)^2 + 4P_0(t)$ is positive at all 8 real roots of $F$, so each root gives two real values of $y$ and there are 16 real points, matching the 16 real roots of $f$. The 8 roots of $F$ lie in two clusters, four in $(0.188965, 0.191162]$ and four in $(0.874512, 0.896484]$; the isolating intervals are listed in Section 4.3.

TABLE 1. Degrees over $\mathbb{Q}$ of the nodes and of the free coefficients. Node degrees were found by integer LLL at 3990 digits and confirmed from the exact data, by factoring the characteristic polynomial of multiplication by each node on $R$ (`node_degrees_exact.py`), which also reproduces the LLL polynomials; the counts of coefficients are the degrees of the minimal polynomials of the 77 free coordinates, computed by the independent checker from the exact data.

| degree | nodes | number of free coefficients |
|---|---|---|
| 1 | $c_1, c_2, c_4, c_5, c_9, c_{10}, c_{11}, c_{13}$ | 0 |
| 2 | $c_8$ | 2 |
| 4 | $c_6, c_7, c_{12}$ | 24 |
| 8 | $c_3$ | 34 |
| 16 | | 17 |

3.4. **Numerical values.** Table 2 gives all 91 coefficients to 30 significant digits at one real embedding of $K$: the one that the numerical search found, with $t_0 = c_3 \approx 0.876959382655$ and $y_0 = c_{12} \approx 0.420844326389$, so that $\theta_0 = t_0 + y_0 \approx 1.2978037090437$ is one of the 16 real roots of $f$. At that embedding the algebraic nodes are

$$c_3 \approx 0.876959382655, \quad c_6 \approx 0.347994313946, \quad c_7 \approx 0.913467098174,$$
$$c_8 \approx 0.145040442433, \quad c_{12} \approx 0.420844326389.$$

Several coefficients are large (for example $a_{13,11} \approx -39.88$, $a_{13,5} \approx 37.62$, $b_5 \approx 6.789$ and $b_{11} \approx -6.544$), which is one more reason to regard the method as an existence proof rather than as a method for practical use (see also Figure 2). The exact values are the polynomials $g_j$ in the data file (Section 8).

3.5. **Structure.** The following properties explain the shape of the method. They were first observed numerically on the component $\Sigma$ (at 120 digits at a point of $\Sigma$), and they hold exactly in $K$ for the certified tableau: a short program using the field arithmetic of the pure-Python checker (`structure_exact_check.py`, 2 s) verifies each equation below and each failure of the next one. They are not used in the proof. In Butcher's notation, $C(\eta)$ at stage $i$ means $\sum_j a_{ij} c_j^{k-1} = c_i^k/k$ for $k = 1, \dots, \eta$, and $D(\zeta)$ at column $j$ means $\sum_i b_i c_i^{k-1} a_{ij} = b_j(1 - c_j^k)/k$ for $k = 1, \dots, \zeta$.

- Every weighted stage other than the first $(5, 6, 7, 8, 10, 11, 12, 13)$ has stage order exactly 3: $C(3)$ holds there and $C(4)$ fails. The unweighted stages $3, 4, 9$ satisfy $C(2)$ and not $C(3)$, and stage 2 only $C(1)$. In particular $a_{3,2} = c_3^2/(2c_2)$.
- $D(1)$ holds in every column. Column 3 satisfies $D(4)$ and not $D(5)$. Columns 2 and 13 satisfy $D(\zeta)$ for every $\zeta$ for trivial reasons ($b_i a_{i,2} = 0$ for all $i$ and $b_2 = 0$; column 13 of $A$ is zero and $c_{13} = 1$). Every other column fails $D(2)$.
- $b$ is the interpolatory quadrature rule of the nine nodes, as shown above.
- $c_4 = c_9 = c_{10}$, $c_{13} = 1$ and $a_{9,2} = a_{4,2} \neq 0$. The coincidences $c_9 = c_4$ and $a_{9,2} = a_{4,2}$ are forced on $\Sigma$ by two identities (Section 5.3).

On Khashin's family $G$, by contrast, only $b_2$ among the weights and $a_{13,2}$ among the coupling coefficients vanish, $C(2)$ holds at stages 3 to 13 with $C(3)$ only at stage 13, $D(1)$ holds and $D(2)$ fails. In neither family does Butcher's classical reduction through $B$, $C$ and $D$ assumptions reduce the 486 conditions to a square system; the reduction that does hold is Khashin's filtered one, which we discuss in Section 5.2.

## 4. The certificate

4.1. **The statement in the ring.** The data file (Section 8) specifies the ring $R$ of Section 3.3 by $F$, $P_0$ and $P_1$, the fourteen fixed rationals (2), and the 77 free coefficients as elements $u(t) + v(t)y$ of $R$. Let $(A, b)$ be the 13-stage tableau with these entries in $R$; it is strictly lower triangular, and $c = A\mathbf{1} \in R^{13}$.

**Theorem 4.1.** *For every rooted tree $t$ with $|t| \le 9$, the element $b\cdot\Phi(t) - 1/\gamma(t)$ of $R$ has reduced form 0.*

Two programs of Section 4.5 verify this statement directly in $R$. The other two verify the same 486 conditions in the univariate form (Section 4.4); that form was converted from the tower form by exact arithmetic, with no further numerics, so they check the same tableau. The recursion for $\Phi$ involves only additions and multiplications in $R$ and rational constants, and the reduced form in the basis $t^i y^j$ is unique, so the statement is decidable by a finite computation and the computation is exactly what the programs do. The proof of Theorem 4.1 is that computation.

4.2. **Specialization.**

**Lemma 4.2.** *Let $R$ be a commutative ring containing $\mathbb{Q}$ and let $(A, b)$ be an $s$-stage tableau with entries in $R$ that satisfies $b \cdot \Phi(t) = 1/\gamma(t)$ in $R$ for every tree with $|t| \le p$. If $\varphi : R \to \mathbb{R}$ is a ring homomorphism, then the real tableau $(\varphi(A), \varphi(b))$ is an explicit Runge-Kutta method of order $p$ with nodes $\varphi(c)$.*

*Proof.* A ring homomorphism fixes $\mathbb{Q}$ and commutes with sums and products, so $\varphi(\Phi(t)) = \Phi_{\varphi(A)}(t)$ componentwise, and $\varphi(b \cdot \Phi(t)) = \varphi(b) \cdot \Phi_{\varphi(A)}(t)$. Applying $\varphi$ to $b \cdot \Phi(t) - 1/\gamma(t) = 0$ gives the real order condition. Strict lower triangularity is preserved, and $\varphi(c) = \varphi(A)\mathbf{1}$. Order $p$ then follows from Butcher's theorem. □

No division occurs anywhere in the recursion, so the lemma needs no field property of $R$: it holds whether or not $F$ is irreducible, and whether or not $R$ is a field. Irreducibility only enters the description of $K$ as a number field of degree 16.

**Lemma 4.3.** *Let $t_0 \in \mathbb{R}$ with $F(t_0) = 0$, and let $y_0 \in \mathbb{R}$ with $y_0^2 = P_1(t_0)\, y_0 + P_0(t_0)$. Then $t \mapsto t_0$, $y \mapsto y_0$ defines a ring homomorphism $R \to \mathbb{R}$. Such $y_0$ exist exactly when $D(t_0) = P_1(t_0)^2 + 4P_0(t_0) \ge 0$, and there are two of them when $D(t_0) > 0$.*

*Proof.* Evaluation $\mathbb{Q}[t, y] \to \mathbb{R}$ at $(t_0, y_0)$ is a ring homomorphism whose kernel contains $F(t)$ and $y^2 - P_1(t)y - P_0(t)$, hence the ideal they generate, so it factors through $R$. The rest is the quadratic formula. □

4.3. **Real roots.**

**Lemma 4.4.** *$F$ is squarefree and has exactly 8 real roots, one in each of the intervals*

$$(0.188965, 0.189102],\ (0.189102, 0.189240],\ (0.189514, 0.190063],\ (0.190063, 0.191162],$$
$$(0.874512, 0.876709],\ (0.876709, 0.878906],\ (0.878906, 0.887695],\ (0.887695, 0.896484],$$

*and $D > 0$ at every one of them. Consequently $R$ has exactly 16 ring homomorphisms to $\mathbb{R}$.*

*Proof.* Sturm's theorem in exact rational arithmetic: the Sturm sequence of $F$ ends in a nonzero constant, so $F$ is squarefree, and the sign-change counts at the endpoints of the intervals above (and at the Cauchy bound for the roots) give exactly one root in each interval and none elsewhere. For the sign of $D$, each isolating interval $[l, h]$ is bisected (keeping the root by Sturm counts) until $\min(D(l), D(h)) - M(h - l) > 0$, where $M$ bounds $|D'|$ on the interval by the sum of the absolute values of the coefficients of $D'$ times the appropriate powers of $\max(|l|, |h|, 1)$; all of this is exact rational arithmetic. Since $R$ is a free module of rank 2 over $\mathbb{Q}[t]/(F)$, every homomorphism $R \to \mathbb{R}$ restricts to one of the 8 real roots of $F$ and sends $y$ to one of the two real roots of the quadratic there. The independent checker also shows this through the univariate form: $f$ has 16 distinct real roots by a Sturm sequence, and $\theta = y + t$ maps the 16 points $(t_0, y_0)$ onto them. □

*Proof of Theorem 1.1.* By Theorem 4.1 the tableau satisfies all 486 order conditions in $R$. By Lemma 4.4 there are 16 ring homomorphisms $\varphi : R \to \mathbb{R}$, and by Lemma 4.2 each gives a real explicit 13-stage method of order 9. For the field statement: $R$ has dimension 16 over $\mathbb{Q}$, the characteristic polynomial of multiplication by $\theta = y + t$ on $R$ is $f$, and $f$ is irreducible, so the minimal polynomial of $\theta$ is $f$ and the subalgebra $\mathbb{Q}[\theta] \cong \mathbb{Q}[T]/(f)$ has dimension 16; hence $R = \mathbb{Q}[\theta]$ is a field, $K = R$ is a number field of degree 16, and the 16 homomorphisms of Lemma 4.4 are its real embeddings, so $K$ is totally real. □

*Proof of Corollary 1.2.* The upper bound is Theorem 1.1. The lower bound $a(9) \geq 12$ is Butcher's theorem $s \geq p + 3$ for $p \geq 8$ [3]. □

4.4. **The univariate check.** The lane's own verifier works in $\mathbb{Q}[T]/(f)$ rather than in $R$. For a tree with $n = |t|$ vertices, the condition $b \cdot \Phi(t) - 1/\gamma(t)$ is a polynomial of degree $n$ in the coefficients, so multiplying it by $f'(T)^n$ and substituting $x_j = g_j(T)/f'(T)$ (with a fixed rational coefficient $r$ entering as $rf'(T)$) gives a polynomial $h_t(T) \in \mathbb{Q}[T]$, which is reduced modulo $f$ and required to vanish. Since $f$ is squarefree, $f'(\theta_0) \neq 0$ at every root $\theta_0$ of $f$, so $h_t \equiv 0 \pmod f$ implies the order condition at every root of $f$. This is the same specialization argument, with $\mathbb{Q}[T]/(f)$ in place of $R$ and the real roots of $f$ in place of the pairs $(t_0, y_0)$. A perturbation of one coefficient of $f$ by $10^{-30}$ makes the check fail (on the toy problem used to validate the pipeline, 3 of 4 conditions failed), so the check is not vacuous.

4.5. **The three checks and their controls.** Table 3 lists the checking programs: two work in the tower form and verify Theorem 4.1, and two work in the univariate form of Section 4.4. They come from three separate sources: the independent checker was written by an agent that did not read the lane's tree, order-condition, tableau or verification code and saw only the layout of the data file; the pure-Python checker was written afterwards to remove FLINT as a shared point of failure and asserts at exit that neither FLINT, SymPy, gmpy2, mpmath nor NumPy was imported. All four read the same data file, which is as it should be: the claim is about that data.

The controls were run with the same code paths:

- Tree counts. The programs of all three sources generate $1, 1, 2, 4, 9, 20, 48, 115, 286$ trees with $1, \ldots, 9$ vertices, in agreement with OEIS A000081, and the independent checker also confirms the extreme densities $\gamma = n$ (bushy tree) and $\gamma = n!$ (tall tree) at every size.
- Classical RK4 over $\mathbb{Q}$: all 8 conditions of orders 1 to 4 hold exactly and all 9 order-5 conditions fail, with residuals $\frac{1}{120}$, $\frac{1}{240}$, $-\frac{1}{240}$, $\frac{1}{120}$, $\frac{1}{80}$, $-\frac{1}{120}$, $-\frac{1}{240}$, $\frac{1}{240}$, $-\frac{1}{120}$ for the trees $[\tau\tau\tau\tau]$, $[\tau\tau[\tau]]$, $[\tau[\tau\tau]]$, $[\tau[[\tau]]]$, $[[\tau][\tau]]$, $[[\tau\tau\tau]]$, $[[\tau[\tau]]]$, $[[[\tau\tau]]]$, $[[[[\tau]]]]$. The same holds for RK4 embedded as constants in $R$.

- Perturbation. Adding $10^{-6}$ to the constant term of $u$ for $a_{3,1}$ leaves orders 1 to 4 intact (because $b_3 = b_4 = 0$ and the stage-order structure absorb the change) and then fails 1 of 9 conditions at order 5, 4 of 20 at order 6, 32 of 48 at order 7, 99 of 115 at order 8 and 270 of 286 at order 9, in all 406 of 486.
- Floating-point pipeline. The high-precision residual code used in the search accepts Verner's 16-stage pair of orders 9 and 8 [25] with residuals between $2 \times 10^{-40}$ and $3 \times 10^{-40}$ at orders 1 to 9 (its coefficients are given to 40 digits) and $1.8 \times 10^{-6}$ at order 10, and it reports DOP853 as order 8 and not 9.

The independent checker also evaluated the exact tableau at the embedding $t_0 \approx 0.876959382655$, $y_0 \approx 0.420844326389$ of Table 2 to 2500 digits and computed the residuals in floating point: at most $8.75 \times 10^{-61}$ at 60 digits and $1.2 \times 10^{-100}$ at 100 digits, which is the rounding floor. This is an observation, not part of the proof.

4.6. **How to rerun the proof.** The pure-Python check needs nothing beyond a Python 3 interpreter. From the folder of the ancillary files (Section 8), run

```
python3 scripts/purepy_exact_check.py results/sigma_rur_inner1_tower.json
```

It prints the tree counts, the RK4 control, the per-order residual counts for the data (all zero), the perturbation control, the Sturm isolation of the 8 real roots of $F$ with the sign of $D$ at each, and a verdict. It ran in 67.0 s with a peak resident size of 23 MB on CPython 3.12.7, and in 64.4 s with 32 MB on CPython 3.15.2 started with `-I -S` (no site packages), with the same output. The FLINT-based checkers need `python-flint` and run in the times of Table 3. A reader with any computer algebra system can also build $A$ and $b$ from the $g_j$ and $f$ of the data file and reduce $b \cdot \Phi(t) - 1/\gamma(t)$ modulo $f$ for the 486 trees.

4.7. **A numerical illustration.** Figure 2 shows the global error of the new method on the Kepler problem against the step size, beside classical RK4 and the order-9 member of Verner's 16-stage pair [25]. Over the four smallest step sizes the local slopes of the new method are 8.85, 8.92 and 8.96, approaching 9; those of Verner's method are 9.54, 9.53, 9.49 and those of RK4 4.44, 4.35, 4.27. At equal step size the error of the new method is $5 \times 10^4$ to $4 \times 10^5$ times that of Verner's method, which is consistent with its large coefficients and with $c_2 = 5/4$: the chart values were chosen to make the algebraic numbers small, not the error constants. A plot of this kind cannot distinguish an exact method from one whose order conditions hold to $10^{-17}$, as Khashin's printed tableau does, which is why the proof is the exact check and not a convergence study.

## 5. How the method was found

Nothing in this section is part of the proof. It records the numerical path that led to the exact data, so that the choices in (2) are not mysterious and so that the method can be reproduced or improved. All computations ran single-threaded on a laptop (Apple M2 Max, 32 GB), in Python with mpmath [22], python-flint (arb and FLINT [21]), NumPy and SciPy, as single-threaded processes, at most two at a time, with a peak resident size of 1.0 GB; the total is roughly 10 to 15 CPU-hours over two days, estimated from the log rather than metered.

5.1. **Refining Khashin's tableau.** Khashin's journal paper [12] is not open access; we worked from his preprint [13], which prints $c_2, \dots, c_{13}$, $b_1, \dots, b_{13}$ and the 78 entries $a_{ij}$ to 17 decimal places, with $b_2 = 0$, $a_{13,2} = 0$ and $c_{13} = 1$ printed as exact. We have not compared these values with the journal version. Two text extractions of the archived PDF agree on all 103 entries.

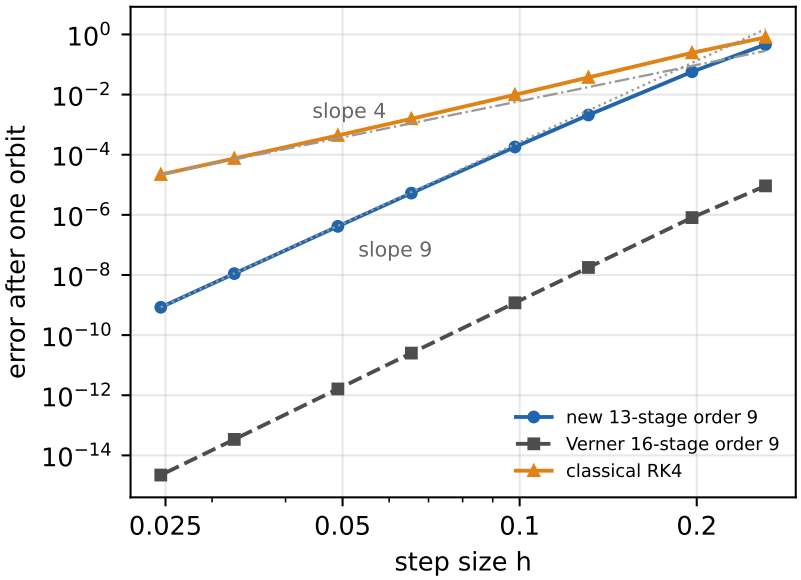



FIGURE 2. A numerical illustration, not part of the proof. The Kepler problem $q'' = -q/|q|^3$ in the plane with eccentricity $e = 1/2$, started at $q = (1-e, 0)$, $q' = (0, \sqrt{(1+e)/(1-e)})$ and integrated over one period $T = 2\pi$ with $N = 24, 32, 48, 64, 96, 128, 192, 256$ constant steps $h = T/N$. The error is the largest component of $|y(T) - y(0)|$ in the four variables $(q, q')$, since the exact solution is periodic. Methods: the new method at the real embedding of Table 2, classical RK4, and the order-9 member of Verner's 16-stage pair of orders 9 and 8 (coefficients to 40 digits). All arithmetic is in mpmath at 60 digits, so rounding plays no part at the plotted error levels. The grey reference lines have slopes 9 and 4 and pass through the smallest-step point of the new method and of RK4.

Read as exact rationals, the printed values give maximal residuals over the trees of each order, at 100 digits, of

$$1.0 \times 10^{-17},\ 3.2 \times 10^{-18},\ 3.3 \times 10^{-18},\ 4.5 \times 10^{-18},\ 7.1 \times 10^{-18},$$
$$9.8 \times 10^{-18},\ 1.3 \times 10^{-17},\ 1.6 \times 10^{-17},\ 2.0 \times 10^{-17}$$

for orders 1 to 9, which is the rounding level of 17-digit coefficients. Minimum-norm Gauss-Newton iteration on the 486 conditions in the 89 unknowns other than $b_2$ and $a_{13,2}$ (exact forward-mode Jacobian in ball arithmetic at 900 bits, SVD pseudo-inverse of rank 84, working precision 250 digits) reduced the maximal residual to

$$1.96 \times 10^{-17},\ 7.0 \times 10^{-31},\ 1.2 \times 10^{-59},\ 8.4 \times 10^{-113},\ 1.1 \times 10^{-223},\ 6.8 \times 10^{-272}$$

in successive iterations, the last being the floor of the working precision; the convergence exponents $\log r_{k+1}/\log r_k$ are $1.81, 1.95, 1.90, 1.99$ (Figure 3). Starting instead from all 91 unknowns, or with the extra equation $c_9 = c_{10}$, or with the first column reset so that each row sums to the printed node, gives the same quadratic convergence. The Jacobian at the refined point has rank 84 in the 89 unknowns (86 in 91), so Khashin's method lies on a smooth five-dimensional family $G$ of 13-stage methods of order 9. The run from all 91 unknowns holds nothing fixed, yet it converges to a point at which $b_2 \approx 2 \times 10^{-292}$ and $a_{13,2} \approx -1.8 \times 10^{-289}$, and at which $c_9 - c_{10}$ and $c_{13} - 1$ vanish to the 250 digits of the working precision; so the relations $b_2 = 0$, $a_{13,2} = 0$,

$c_9 = c_{10}$ and $c_{13} = 1$ hold at the refined point although that run did not impose them. The refined tableau is in the data deposit (Section 8); we stress again that this is numerical evidence only.

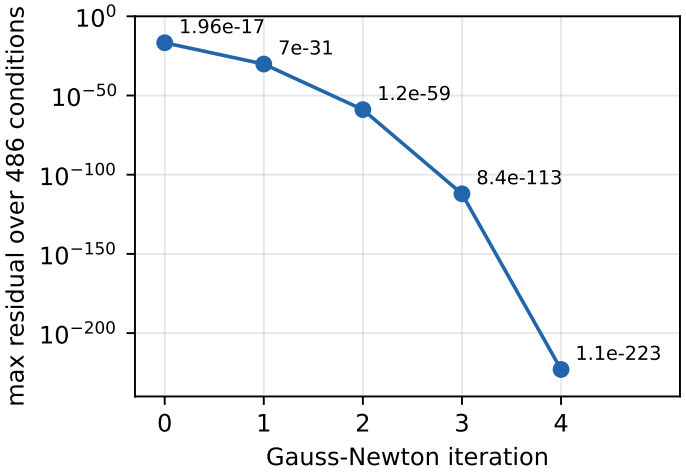



FIGURE 3. Gauss-Newton refinement of Khashin's printed tableau: the maximal residual over the 486 order conditions against the iteration number, on a logarithmic scale. The start is the printed 17-digit tableau read as exact rationals with $b_2 = a_{13,2} = 0$ held (89 unknowns); each step is the minimum-norm Gauss-Newton step with the SVD pseudo-inverse of rank 84, at a working precision of 250 digits. The residuals at iterations 0 to 4 are $1.96 \times 10^{-17}$, $7.0 \times 10^{-31}$, $1.2 \times 10^{-59}$, $8.4 \times 10^{-113}$ and $1.1 \times 10^{-223}$: the number of correct digits roughly doubles at every step, with exponents $\log r_{k+1} / \log r_k$ of 1.81, 1.95, 1.90 and 1.99. The next iterate, $6.8 \times 10^{-272}$, sits at the floor of the working precision and is not plotted. Quadratic convergence of this kind is what Gauss-Newton gives at a solution where the Jacobian has constant rank; it is strong evidence, not proof, that Khashin's method is of order 9.

5.2. **Why not a square system and an interval test.** The standard way to certify a numerical solution of a polynomial system is an interval Newton or Krawczyk test [15] on a square system. The 486 conditions in 89 unknowns are far from square, and a reduction to a square system must come with a proof that its solutions satisfy all 486 conditions. Butcher's $B$, $C$, $D$ assumptions do not give one here (Section 3.5). What does hold on $G$ is Khashin's filtered structure [12, 11]: with $q_m = Ac^m - c^{m+1}/(m+1)$, the spaces $W_2 = \mathrm{span}(q_1)$ and $W_{k+1} = W_k + AW_k + c.W_k + \sum_{i+j=k+1} W_i.W_j + \mathrm{span}(q_k)$ (products componentwise) have dimensions $1, 2, 4, 6, 8, 10, 11, 12$ for $k = 2, \ldots, 9$ at the refined point, and $b$ annihilates $W_8$ to $2 \times 10^{-246}$. We expect that an induction on $k$ shows that the inclusions defining the $W_k$, the hypothesis $b\,W_8 = 0$ and the quadrature conditions $B(9)$ together imply all 486 conditions, but we have only a sketch of that argument, and nothing in this paper depends on it. The mechanism appears to be that of Stepanov's $Q$-type conditions [20] and of the sufficiency theorem of He and Huang [9]; we have not checked that their hypotheses match ours. Written as polynomial equations, however, the structure is overdetermined: about 102 equations against 86 local constraints, with 16 redundant ones. A Krawczyk test on a square subsystem of 291 equations did certify a unique zero in a box of radius $10^{-200}$ around the refined point (contraction ratio

$4.8\times10^{-43}$ at 1000 bits), but five scalar conditions, equivalent to $B(5)$, $B(8)$, $B(9)$, $b\,C^2A^4e_2=0$ and $b\,Cq_6=0$ with $C=\mathrm{diag}(c)$, could not be shown to follow from the kept equations, and so this route produced no certificate. The remaining route was an exact algebraic point, which needs a point of the solution set of small degree over $\mathbb{Q}$. Fixing five coupling coefficients of $G$ at rationals with denominators up to 1000 gave a point (polished to $8\times10^{-997}$ at 1000 digits) at which $c_3$, $c_4$, $c_7$ and $b_1$ satisfy no integer polynomial of degree at most 12 within the reach of that precision, and monodromy loops on such a chart of $G$ found at least 13 points in the fibre without saturating, so $G$ itself looked unpromising for this route.

5.3. **The component $\Sigma$.** Continuing along $G$ toward the hypersurface $b_3=0$ (at 120 digits, holding the 486 conditions) reached a point $P$ at which ten coordinates vanish to $10^{-107}$ or less: $b_3$ itself, $b_4$, $b_9$ and $a_{i,2}$ for $i=5,6,7,8,10,11,12$. The coordinate $a_{9,2}$ does not vanish; it equals $a_{4,2}$. At $P$ the Jacobian of the 486 conditions has rank 84 in the 89 unknowns and rank 74 in the 79 unknowns that remain when the ten vanishing coordinates are removed, both with nullity 5, and the ten coordinates are stationary along the whole five-dimensional tangent space. So near $P$ the solution set is a smooth five-manifold on which the ten coordinates vanish identically: $P$ lies on a second five-dimensional component $\Sigma$, not on $G$ (on $G$ they do not vanish near Khashin's point; at $x^*$, for example, $b_4\approx-6.4\times10^{-3}$, $b_9\approx1.1\times10^{-2}$ and $b_3\approx6.7\times10^{-5}$). The two components meet in a four-dimensional junction, where on $G$ the coordinates $b_4$, $b_9$ and $a_{i,2}$ vanish to first order and $b_3$ to second order ($b_3/b_4^2\approx0.86$ and $b_9/b_4\approx-1.00$ near the junction), which is why the continuation toward $b_3=0$ became stiff and ended on $\Sigma$. Figure 4 is a schematic of this picture.

On $\Sigma$, $c_4=c_9=c_{10}$ and $c_{13}=1$, every weighted stage has stage order 3, and $b$ is the quadrature rule of the nine distinct nodes (Section 3.5). Two of the coincidences are explained by exact identities under the structural hypotheses (the vanishing of column 2 outside rows 3, 4, 9, and the pivot structure of the filtration): the relation "$C^2u\in W_5$" with $u=Ae_2$ has residual exactly $a_{9,2}(c_9-c_3)(c_9-c_4)$ at row 9 and zero at the weighted rows, and, once $c_9=c_4$, the relation "$u.u\in W_6$" has residual $a_{9,2}(a_{9,2}-a_{4,2})$ at row 9. On the chart $a_{9,2}\neq0$, $c_9\neq c_3$ this forces $c_9=c_4$ and then $a_{9,2}=a_{4,2}$; both identities were checked numerically at perturbed points off the coincidence locus to about $10^{-103}$.

The structure on $\Sigma$ is no more reducible to a provable square system than on $G$ (the count of unproved scalar conditions is again five). What $\Sigma$ offers is fewer unknowns and more rational structure, and so the hope of a point of small degree.

5.4. **Charts, fibre degrees and the inner nodes.** A chart is a choice of five coordinates to hold at rational values; the fibre over a chart value is a finite set of points of $\Sigma$, and the point found numerically is algebraic of degree at most the size of its Galois orbit in that fibre. Monodromy loops around a rational base point of the coordinate chart $(a_{2,1},a_{13,6},a_{13,5},a_{13,9},a_{12,5})$ found $15,21,27,33,39,45,51,57$ distinct fibre points after successive runs without saturating, so that chart's degree is at least 57. A chart of nodes, $(c_2,c_6,c_{11},c_{12},a_{13,9})=(5/4,1/3,3/4,1/2,-4)$, gave a point at which $c_4$ satisfies an irreducible integer polynomial of degree exactly 29 (found by LLL [16] at 1000 digits and vanishing beyond the digits used in the fit, with 9 real roots), while $c_5$ and $c_7$ are conjugate over $\mathbb{Q}(c_4)$, so that point's field has degree at least 58.

The decisive observation came from the conditioning of charts. On the tangent space of $\Sigma$ at that point, the charts $(c_2,c_4,c_8,c_5,a_{13,9})$ and $(c_2,c_4,c_5,c_7,a_{13,9})$ have smallest singular values $3.7\times10^{-14}$ and $2.3\times10^{-15}$, against $4.1\times10^{-5}$ for $(c_2,c_4,c_5,c_{11},a_{13,9})$ and $5.3\times10^{-5}$ for the node chart. So $dc_7$ and $dc_8$ are combinations of $dc_4$ and $dc_5$ along $\Sigma$: the inner nodes move on a two-dimensional surface of their own, independently of $c_{11}$ and $c_{12}$. The same holds for $c_6$: at the certified point the charts $(c_2,c_4,c_5,c_6,a_{13,9})$ and $(c_2,c_4,c_8,c_6,a_{13,9})$ have smallest singular values $1.3\times10^{-14}$ and $9.5\times10^{-16}$, against $2.3\times10^{-5}$ for the certified chart. So the inner nodes

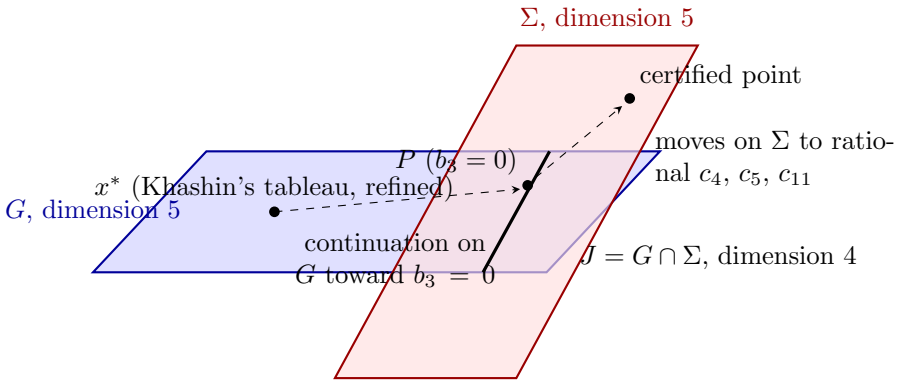



FIGURE 4. Schematic, not to scale, of the solution set of the 486 order conditions near the two families; the dimensions are those of the real solution set, both Jacobians having nullity 5. Khashin's refined tableau $x^*$ lies on a smooth five-dimensional component $G$, on which only $b_2$ among the weights and $a_{13,2}$ among the coupling coefficients vanish. A continuation along $G$ toward $b_3 = 0$ becomes stiff near the junction and ends at a point $P$ of a second five-dimensional component $\Sigma$, on which $b_3 = b_4 = b_9 = 0$ and $a_{i,2} = 0$ for $i \in \{5, 6, 7, 8, 10, 11, 12, 13\}$ as well. $G$ and $\Sigma$ meet in a four-dimensional junction $J$; on $G$ near $J$ the coordinates $b_4$, $b_9$ and $a_{i,2}$ vanish to first order and $b_3$ to second order, and $P$ lies on $\Sigma$ at a distance of about $3 \times 10^{-9}$ from $J$. Moving on $\Sigma$ to the chart values $(c_2, c_4, c_5, c_{11}, a_{13,9}) = (5/4, 6/25, 187/250, 3/4, -4)$ gives the certified point. It and its 15 Galois conjugates are 16 real points of the fibre, over these chart values, of the locus cut out by the 486 conditions and the zeros of $\Sigma$; whether the conjugates lie on $\Sigma$ itself was not checked. None of the 13 fibre points found by monodromy on a chart of $G$ lies on $\Sigma$.

are $c_4, \ldots, c_8$, and if $c_4$ and $c_5$ are held rational, $c_6$, $c_7$ and $c_8$ become algebraic over $\mathbb{Q}$ of small degree whatever the outer nodes do.

Accordingly the chart $(c_2, c_4, c_5, c_{11}, a_{13,9}) = (5/4, 6/25, 187/250, 3/4, -4)$ was chosen and reached from the node-chart point by a real path of 286 continuation steps (a first target, $(c_4, c_5) = (1/4, 37/50)$, met a fold 3 percent of the way along, and a complex detour ended at a non-real point). Newton's method at 130 digits converged to a residual of $1.4 \times 10^{-141}$, confirmed by an independent evaluation at $4.5 \times 10^{-135}$. The point was then refined by a chord iteration with a frozen pseudo-inverse to 4000 digits ($9.4 \times 10^{-4005}$, 23 s), 10000 digits ($4.5 \times 10^{-10005}$, 200 s, 406 MB) and 25000 digits ($1.05 \times 10^{-25004}$ on all 486 conditions, 1737 s, 647 MB).

5.5. **Reconstruction.** Integer LLL at 3990 digits (fitting 3800 digits and checking the remainder) found minimal polynomials of degree 2 for $c_8$ (coefficients of 13 digits), degree 4 for $c_7$ and $c_6$ (19 digits), $c_{12}$ (75 digits), $b_1$ and $b_5$ (149 and 181 digits) and $a_{13,12}$ (597 digits), and degree 8 for $c_3$ and $a_{3,1}$ (72 and 147 digits); each relation vanishes beyond the digits used in the fit, and nothing was found at lower degree. The octic $F$ of $c_3$ is irreducible over $\mathbb{Q}$ with 8 real roots. Over $\mathbb{Q}(c_3)$, 31 of the 77 free coordinates were found with the same polynomials at 3990 and at 9990 digits (heights up to 341 digits), and $c_{12}$ is quadratic over $\mathbb{Q}(c_3)$, which suggested $K = \mathbb{Q}(c_3, c_{12})$. At the 25000-digit point, LLL in the basis $c_3^i, c_3^i c_{12}$ $(0 \le i \le 7)$ found every one of the 77 free coordinates in the form $u(c_3) + v(c_3)c_{12}$; the integer relations had heights of up to 841 digits, so ten coordinates needed a fit of 19600 digits, and every relation also vanishes in the digits the fit did not use, reproducing each coordinate to $10^{-24988}$. This took 132 s. The

tower was then converted exactly (no further numerics) to the primitive element $\theta = c_{12} + c_3$: $f$ is monic, squarefree, of degree 16, with rational coefficients (numerators of up to 434 digits, denominators of up to 430), and every coordinate is $g_j(\theta)/f'(\theta)$ with $g_j \in \mathbb{Q}[T]$.

A lattice reconstruction is a guess until it is checked. The exact check of Section 4 is that check; an error in any reconstructed coordinate would appear as a nonzero residual, as the perturbation control shows. The exact verification, the real-root isolation and a reconstruction from a complete fibre were validated first on a toy problem, the two-parameter family of 3-stage methods of order 3 with two coordinates fixed, whose complete fibre of 4 points and eliminant are known exactly from a Gröbner basis; there the reconstructed $f$ and $g_j$ agree with the Gröbner basis coefficient by coefficient, and dropping a fibre point makes the reconstruction fail outright. The toy does not exercise the single-point LLL reconstruction used for $\Sigma$, which had its own controls: the minimal-polynomial search finds the degree-6 polynomial of $\sqrt{2}+3^{1/3}$ and nothing for a generic number up to degree 30, and the relation search recovers planted relations with coefficients of 1, 41 and 91 digits.

## 6. Discussion

6.1. **Khashin's method and the two components.** Our theorem does not certify Khashin's tableau. The evidence for it is the quadratic Gauss-Newton convergence to $1.1 \times 10^{-223}$ and the smooth five-dimensional family through it; an exact certificate for a point of $G$ would need a chart of $G$ with a fibre of moderate degree, which we did not find. The component $\Sigma$ is new as far as we know. Its pattern of zeros is in the spirit of Khashin's families: among his four families of 11-stage methods of order 8 he lists the case $b_2 = b_3 = b_4 = a_{11,2} = a_{10,2} = \cdots = a_{5,2} = 0$, of local dimension 7 [13, Section 4.3], and $\Sigma$ has the analogous zeros at 13 stages, with $b_9 = 0$ and $a_{9,2} = a_{4,2} \neq 0$ in place of a vanishing $a_{9,2}$.

6.2. **Simpler fields.** The degree 16 and the coefficient sizes are properties of the particular rational chart values, not of $\Sigma$. The quadratic tower of Section 3.3 shows where the degree comes from: with $c_4$ and $c_5$ rational, $c_8$ is quadratic over $\mathbb{Q}$, $c_6$ and $c_7$ are quadratic over $\mathbb{Q}(c_8)$, $c_3$ is quadratic over $\mathbb{Q}(c_7)$, and $c_{12}$ is quadratic over $\mathbb{Q}(c_8)$ but not in $\mathbb{Q}(c_3)$. We tried three variations, each on a point refined to 4000 digits and scanned by LLL at 3990 digits, so the results are numerical. Holding $(c_4, c_8) = (6/25, 29/200)$ rational instead makes $c_5$, $c_6$ and $c_7$ the three roots of one irreducible cubic and gives $c_3$ degree 12, so a field of degree at least 24. Holding $(c_5, c_8) = (187/250, 29/200)$ gives $c_4$ degree 4 and $c_6$, $c_7$ degree 8. Moving $c_{11}$ from $3/4$ to $4/5$ or $5/6$ leaves the minimal polynomials of $c_3$, $c_7$ and $c_8$ unchanged and changes the quartic of $c_{12}$, which stays quadratic over $\mathbb{Q}(c_3)$. So the inner nodes set the degree, and a smaller field needs inner nodes at which more of the tower collapses; along $c_4 = 6/25$, fits of $c_8$'s conjugates as rational functions of $c_5$ suggest that $c_8$ is rational only at the rational points of a curve of genus 2. A tableau short enough to print, perhaps in nested square roots, would be the natural follow-up. The fibre degree of the inner chart over $\mathbb{Q}$ is not known; the 16 roots of $f$ are one Galois orbit in it.

6.3. **Twelve stages.** Whether $a(9) = 12$ is open. A proof of nonexistence in Butcher's style would need an argument of the kind that proves $s \geq p + 3$ for $p \geq 8$ [3], and we have none. A heuristic from the filtered structure points to 13: in the refined Khashin method, in Stepanov's 15-stage order-10 method and in Zhang's 16-stage order-10 method, the filtration satisfies $\dim W_{p-1} = s - 2$, $\dim W_p = s - 1$ and $b\, W_{p-1} = 0$, and the graded dimensions of the minimal-stage methods of orders 6 to 9 follow the pattern $1, 1, 2, \ldots, 2, 1, 1$ with $p - 5$ twos (we computed this for the order-9 method; for orders 6 to 8 we take it from Khashin's 2016 seminar slides [14]), which gives $s = 2p - 5$: $7, 9, 11, 13$, and 15 for $p = 10$, which Stepanov reached. A

12-stage method of order 9, or a 14-stage method of order 10, would have to break this pattern at some level. No theorem forbids that.

6.4. **Order 10.** The same exact-point route applies in principle to the 15-stage order-10 family of Stepanov, whose coefficients are already exact, and to Zhang's 16-stage method, which is known only numerically [28]. For the latter the open question is the same as it was for Khashin's method, and the answer may again be a different component rather than the point that was found.

## 7. Use of AI tools

The searches, the numerical computations, the structural analysis and the checking programs described in this paper were carried out by AI agents (Anthropic's Claude models) under my direction, over two days. I set the goals, the pass and fail criteria, which were frozen in writing before any computation on Khashin's tableau, and the compute limits. Ideation memos from a second model family (OpenAI's GPT-6 Astra) were used in planning the certificate route and contributed no part of the proof. I take responsibility for the result. The correctness of Theorem 1.1 rests on the exact check of Section 4, which a reader can rerun in about a minute without trusting any of these tools; the controls in Section 4.5 show that the check is not vacuous.

## 8. Data availability

The exact data are in the file `results/sigma_rur_inner1_tower.json` (2.3 MB), whose SHA-256 digest is

`eb10c24c6c683841f4b920dcc1b84fae4bc6fac6eaf2066c5096de471bc859e3`.

It contains the fixed and free coordinates, the tower form ($F$, $P_0$, $P_1$ and the pairs $(u, v)$ for every free coordinate) and the univariate form ($f$ and the $g_j$). The checking scripts `purepy_exact_check.py`, `fibre_verify.py` (with `fibre_common.py`) and `indep_exact_core.py`, `indep_exact_tower.py`, `indep_exact_univariate.py`, `indep_exact_field.py`, `indep_exact_controls.py`, `structure_exact_check.py`, `node_degrees_exact.py`, `subfield_test.py`, the 30-digit tableau at each of the 16 real roots, the certificate file with the isolating intervals, and the refined Khashin tableau are included with this paper as ancillary files on arXiv, with a README and a list of SHA-256 digests, and are also available at `https://github.com/doubleMcyber/rk13-order9` [17].

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TABLE 2. The 91 coefficients to 30 significant digits at the real embedding $c_3 \approx 0.8769593827$, $c_{12} \approx 0.4208443264$ (the real root $\theta_0 \approx 1.2978$ of $f$). The exact values are $g_j(\theta_0)/f'(\theta_0)$, or equivalently $u_j(t_0) + v_j(t_0)\, y_0$ in the tower form.

| | | | |
|---|---|---|---|
| $a_{2,1}$ | $5/4$ | $a_{11,2}$ | $0$ |
| $a_{3,1}$ | $0.569336279124272835419896600 78$ | $a_{11,3}$ | $0.126084740303424175479440715292$ |
| $a_{3,2}$ | $0.307623103530486980813374703343$ | $a_{11,4}$ | $0.719409958129716852818174761612$ |
| $a_{4,1}$ | $0.231197094669675835846238563328$ | $a_{11,5}$ | $8.791182012820643624204572581 86 \times 10^{-3}$ |
| $a_{4,2}$ | $0.056509153684038099972554076 7305$ | $a_{11,6}$ | $-0.028598418003687807565705347794$ |
| $a_{4,3}$ | $-0.047706248353713935818792640 0582$ | $a_{11,7}$ | $4.229641043114283747486022161 28 \times 10^{-5}$ |
| $a_{5,1}$ | $-0.073819817163238223702768296 6135$ | $a_{11,8}$ | $-0.029324032311204721241361460 7061$ |
| $a_{5,2}$ | $0$ | $a_{11,9}$ | $-0.012737515247596451548809048 8057$ |
| $a_{5,3}$ | $0.129545534814026203860300406886$ | $a_{11,10}$ | $0.036117644325755619292290088 4566$ |
| $a_{5,4}$ | $0.692274282349212019842467889727$ | $a_{12,1}$ | $0.195605760210684955018831033787$ |
| $a_{6,1}$ | $0.235352993684173820542449787131$ | $a_{12,2}$ | $0$ |
| $a_{6,2}$ | $0$ | $a_{12,3}$ | $-0.201503084933281010125465892 37$ |
| $a_{6,3}$ | $-0.311104068558553122363714154933$ | $a_{12,4}$ | $-0.288958466796165325014050207201$ |
| $a_{6,4}$ | $-0.032311225071654238362578510192 1$ | $a_{12,5}$ | $4.198700664035656168330518229 88$ |
| $a_{6,5}$ | $0.456056613891589431387001610463$ | $a_{12,6}$ | $0.282686952385962904426880706087$ |
| $a_{7,1}$ | $0.326488231648316077434783459533$ | $a_{12,7}$ | $5.852545906116497425845816586 97 \times 10^{-3}$ |
| $a_{7,2}$ | $0$ | $a_{12,8}$ | $0.172475401629383326028309006679$ |
| $a_{7,3}$ | $-0.316781663393330347354359987568$ | $a_{12,9}$ | $-0.029013456673248275825259091 7818$ |
| $a_{7,4}$ | $-0.524758885672715491300168853812$ | $a_{12,10}$ | $-0.016600358265274869802235379 9384$ |
| $a_{7,5}$ | $0.809591542204789093557608452467$ | $a_{12,11}$ | $-3.898401631110895070155480565 14$ |
| $a_{7,6}$ | $0.618927873386629134862501768521$ | $a_{13,1}$ | $-0.781460808800558575954099010456$ |
| $a_{8,1}$ | $0.154969390655413123449502349834$ | $a_{13,2}$ | $0$ |
| $a_{8,2}$ | $0$ | $a_{13,3}$ | $1.305902226517924598005402770 21$ |
| $a_{8,3}$ | $-0.152192184389734769520240489255$ | $a_{13,4}$ | $7.000496458395378023583049473 4$ |
| $a_{8,4}$ | $-0.044593857837301699160704633335 3$ | $a_{13,5}$ | $37.615793452336358485256066918 1$ |
| $a_{8,5}$ | $0.228189723691185435625908783746$ | $a_{13,6}$ | $-14.423690853926520972104854189 7$ |
| $a_{8,6}$ | $-0.038476367925023634080793798656 2$ | $a_{13,7}$ | $0.661052965762676360013390477514$ |
| $a_{8,7}$ | $-2.856261761604219803385996997 92 \times 10^{-3}$ | $a_{13,8}$ | $-3.301643378313321883285237675 41$ |
| $a_{9,1}$ | $1.986071321334486258999623687 61$ | $a_{13,9}$ | $-4$ |
| $a_{9,2}$ | $0.056509153684038099972554076 7305$ | $a_{13,10}$ | $9.785392824699894420472589963 56$ |
| $a_{9,3}$ | $-2.537029871657742049980924707 95$ | $a_{13,11}$ | $-39.878838416706264395348225493$ |
| $a_{9,4}$ | $-1.910077313725402469345421865 04$ | $a_{13,12}$ | $7.016995530034433939361916765 82$ |
| $a_{9,5}$ | $3.701878525321469563123145579 43$ | $b_1$ | $0.040112567074549427654578060 84$ |
| $a_{9,6}$ | $0.451674511641078763673906113 79$ | $b_2$ | $0$ |
| $a_{9,7}$ | $-0.085709142162416325692202775 2575$ | $b_3$ | $0$ |
| $a_{9,8}$ | $-1.423317184435511840750680109 3$ | $b_4$ | $0$ |
| $a_{10,1}$ | $0.913751353497007830991796021676$ | $b_5$ | $6.789328425534730112133668801 66$ |
| $a_{10,2}$ | $0$ | $b_6$ | $0.119401156766443208066080941456$ |
| $a_{10,3}$ | $-1.296336841838013092816336338 91$ | $b_7$ | $0.168853411780417316559823581192$ |
| $a_{10,4}$ | $-0.614725754402159405639176720486$ | $b_8$ | $0.253829509287975506589677546062$ |
| $a_{10,5}$ | $1.920311536437142408806184875 99$ | $b_9$ | $0$ |
| $a_{10,6}$ | $-0.033994399240438636099657976 0337$ | $b_{10}$ | $-0.074149515896722387230522417726 7$ |
| $a_{10,7}$ | $-0.025181292242267744795264513 1598$ | $b_{11}$ | $-6.543687414001227737826514790 15$ |
| $a_{10,8}$ | $-0.645811269858641809693990245357$ | $b_{12}$ | $0.227532229210045117715169951252$ |
| $a_{10,9}$ | $0.021986667647370449246444896 2741$ | $b_{13}$ | $0.018779630243789436338038325 4179$ |
| $a_{11,1}$ | $-0.069785855619659453695709140 8579$ | | |

TABLE 3. The exact checks of the 486 conditions. Times are single-thread wall times, on the laptop described in Section 5, of the 486-condition check alone; the lane's verifier, which does nothing else, was run in two chunks of trees. The whole runs, with controls and root isolation, took 67.0 s (pure Python, CPython 3.12.7; 64.4 s on 3.15.2), 145.7 s (independent tower check) and 2197.8 s (independent univariate check, almost all of it an exact sweep over the 16 roots).

| program | arithmetic | form | time |
|---|---|---|---|
| `fibre_verify.py` (lane) | FLINT `fmpq_poly` mod $f$ | univariate | 20.9 s + 62.4 s |
| `indep_exact_tower.py` (indep.) | FLINT `fmpq_poly` in $R$ | tower | 1.5 s |
| `indep_exact_univariate.py` (indep.) | FLINT, $f'$ inverted by xgcd | univariate | 38.8 s |
| `purepy_exact_check.py` | Python `int`, `fractions` | tower | 21.5 s |